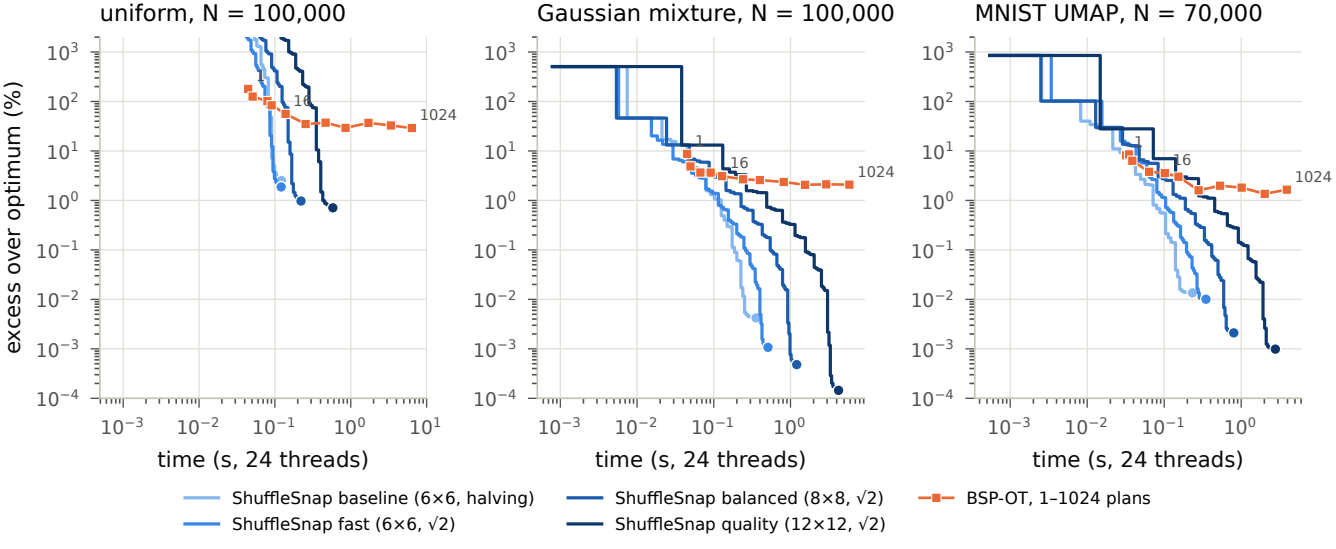
uniform, N = 100,000
Gaussian mixture, N = 100,000
MNIST UMAP, N = 70,000
excess over optimum (%)
time (s, 24 threads)
1
16
1024
ShuffleSnap baseline (6×6, halving)
ShuffleSnap fast (6×6, √2)
ShuffleSnap balanced (8×8, √2)
ShuffleSnap quality (12×12, √2)
BSP-OT, 1–1024 plans